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INTERNAL ASSIGNMENT ANSWER SHEET

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NAME	**DATA VISUALIZATION**

Assignment SET–1

Question 1

Answer:

Concept and significance

Creating a visual with data is a great way to take a pile of boring numbers and give it to our brains in a much more digestible form. Creating a visual removes the guesswork from the data,, allowing it to be presented as something that's easily understood. Our brains tend to be more visual, which allows us not to be overwhelmed by all the data we have to deal with. It can help expose hidden patterns,, identify errors, and make intelligent decisions.

Benefits of visualizing data in different industries:

1. **Healthcare**: Doctors are given limited time to review extensive lab reports. Therefore presenting critical patient data visually will help physicians easily detect any significant changes in their patient's health.
2. **Business**: Transform raw sales data into a visually intuitive dashboard to give yourself a quick idea of what products are your best sellers.
3. **Finance**: A simple line graph will help you visually identify any large fluctuations among more complex financial data.
4. **Marketing**: Visualizing click through tracking will provide you with immediate evidence of which advertisements are generating clicks. And which marketing campaigns are currently earning a return on investment.

Four General Principles of Creating Effective Data Visualizations:

1. **Clarity**: Your audience shouldn't have to play detective to figure out what you mean by the chart you created. You want your titles, labels and axes to be as obvious to your audience at first glance as if they haven't been created by you.

2. **Simplicity**: Less is more; eliminate distractions. Use fewer colors; use a simple title; eliminate 3D charts.
3. **Accuracy:** don't ever, ever manipulate the data for the sake of making your chart look more exciting! Create and maintain truthful scales for your axes, so you won't accidentally deceive your audience.
4. **Storytelling:** A good chart isn't just an image; it's a story. The chart should also assist in directing the viewer's eye to the crucial point of the chart (using highlights and short annotations), so the viewer understands what happened and why it matters.

Significance of choosing the right chart

Even accurate information can confuse people if it is presented the way. An incorrect picture changes the meaning. Covers up the important point. Different charts are made for purposes.

1. I find that Line charts are best for tracking trends over time.
2. Bar charts are useful when comparing categories.
3. Scatter plots show relationships between two variables.
4. Pie chart is, for showing a few parts of a whole.
5. Histograms show how data is distributed.

In conclusion, Data Visualization is much more than a decoration; it is a communication tool that turns data into understanding. When a visual is clear, simple , and built around story, and when the right chart is chosen, people in any field can make faster and smarter decisions.

Question 2

Answer:

The role of Microsoft Excel in Data Visualization and Analysis

Microsoft Excel is used by many different types of workers throughout India to help organize, analyze and present data. The use of rows and columns provides an easy way to store large amounts of data. Formulas, filters, sorting and PivotTables provide ways to take unorganized, raw numbers and create useful information from them. In addition to using formulas and tables to analyze data, Excel also has built-in visualizations (Charts) that are automatically updated as you enter new data, this allows managers to see trends in their company's performance, with no need for special software. This makes it a very popular tool among analysts, finance departments, and students due to its wide availability within companies and ease of learning.

Data Type

Excel works with kinds of data. These include text, like names, cities or product codes. It also handles numbers, such as sales, quantity and price. You can use date and time values too. Currency

and percentage formats are common well.. There are logical values, which are either True or False. Using the data type matters a lot. Formulas and charts rely on it. If you put a date in a text format for example it won't show up correctly on a time axis. The chart might ignore it. Mess up the order. So always check how your data is stored.. Make sure dates are as dates, numbers,, as numbers and so on. This helps everything work smoothly.

Data Importing and File Management

You can enter information manually. Import it from CSV file or, from databases. The web using the Data tab.. When you have imported it make sure to clean it up by removing duplicates fixing errors and ensuring the formats are correct. To stay organized save your work in a file with a clear and easy to understand name.. Make copies protect information with a password and always keep the original data on a separate sheet to avoid mistakes.

Creating and Formatting charts

To create a chart first highlight the data you want then go to the Insert tab and choose the chart type that fits best.:

1. **Bar chart:** Bar charts are great for comparing categories, such as regional sales. Just make sure the chart's axis starts at zero, Try sorting the bars so the differences are clear at a glance.
2. **Line chart:** The line chart is used to track trends over time such as revenue.. The chart's timeline should always run across the horizontal axis.
3. **Pie chart**: The pie chart shows each part's share of the market. It works best with four or five categories that add, up to 100 percent.

Formatting is done through the chart design and format tabs. A clear title helps. Labelled axes are good. Data labels are useful. A consistent color scheme keeps things tidy. Light gridlines add clarity. A placed legend makes a chart easier to understand. Unnecessary effects such, as 3D should be avoided. They distort the values of a chart.

Factor in Selecting an Appropriate chart

1. **Purpose**: I always check if the chart is meant to compare trends show distribution reveal relationships or illustrate composition.
2. **Type of data:** Is the data time-based or numerical?
3. **Volume of data:** I notice that if there are too many categories, bar and pie charts become crowded and hard to read.
4. **Audience:** I see that senior management likes visuals,, with a clear message. Technical users can handle details.,
5. **Key message**: I make sure the chart highlights one point.
6. **Accuracy:** I always ensure scales and proportions do not mislead the viewer.

In conclusion, Excel is a practical and accessible tool for both analysis and visualization. When data type is correct, files are well managed, charts are properly formatted and chart type suits the purpose and audience, Excel helps professionals present information clearly and make better decision

Question 3

Answer:

Advanced Data Analysis and Visualization Features in Excel

Formulas and charts, in Excel offers me some advanced tools that help manager analyze large dataset quickly and show insights in a way that is interactive and easy to understand. These Excel features reduce effort and support faster data driven business decisions.

Pivot Tables

Pivot Table help us summarizes datasets without writing formulas. By moving fields into Rows, Columns, Values and filters the manager can look on sales, region, by product or by month in seconds. Pivot Table help us in sum,, count., average or show percentages and the layout can be rearranged instantly. Pivot Table significance lies in turning thousands of transaction records into a report that answers specific business questions.

Pivot Charts

A pivot chart is a chart linked directly to a Pivot table.. When the table is filtered or sorted,, the chart updates automatically. This makes it useful for presentations as the same data can be shown from different angles without rebuilding the chart.

Conditional Formatting

Conditional Formatting helps in changes in the appearance of cells based on rules. Color scales, data bars and icon sets highlight high and low values; targets met or missed, overdue payments or unusual figures. Managers can spot these problem areas briefly,. which makes it valuable for performance tracking and exception reporting.

Sparklines

Sparklines are like tiny charts placed inside a single cell, in line, column. show win and loss patterns. Sparklines let you see the trend of a row of data, such as monthly sales for each branch

without having a need big chart. Sparklines save space. Are ideal for dashboards and summary tables.

Advanced Chart Formatting

Excel chart formatting allows detailed control over charts; you can include combination charts (for example, column revenue and a line for profit margin) secondary axes, trendline, data labels, error bars, custom colors and chart templates. Proper formatting highlights the key message, improves readability , and gives reports a professional finish.

Slicers

Slicers are filter buttons that connected to pivot tables and pivot charts. Instead of navigating drop down menus, user clicks on a region, product, or category to filter the data instantly. Only one slicer can control several tables and charts at once, which makes reports amazingly interactive and easy for non technical users.

Timelines

A timeline is a filter for date fields. A timeline helps users filter data by days, months, quarters, or year by dragging a bar along a time scale. Timelines are especially useful for analyzing seasonal patterns and comparing performances across periods.

Significance for Business

These tools together allow analysts to build dashboards that update as the data changes. These tools save time reduce errors improve clarity and let decision makers explore the data themselves rather than relying on some other static reports. In areas such as sales,, finance, HR, and operations, this leads to better informed decisions.

Conclusion

Pivot Tables, Pivot Charts, Conditional Formatting, Sparklines, Advanced Chart formatting, Slicers and Timelines make Excel a strong business analytics tool. These tools used together turn messy data into clear, interactive, and actionable insights.

Assignment SET–2

Question 4

Answer:

Tableau is a business intelligence and data visualization tool. It helps the users explore data and build interactive visuals through a drag and drop interface. The best part of tableau is that we can

do it without writing any code. This tool is widely used by organizations to convert large datasets into meaningful insights where stakeholders and managers can act on.

Tableau Environment and key features

In Tableau the workspace includes a data pane. This pane shows fields as dimensions and measures. Dimensions are categories like region or product. Measures are numbers like sales or profit. Users drag these fields onto the rows and columns shelves. The marks card controls how data looks. It changes colors, size, labels and tooltips. The Show Me panel suggests chart types based on the data. This makes it easier to pick the visual..

Some of the features include are fast drag and drop analysis. It offers the user filter to quickly change what they see. Calculated field help the user create Data values from existing ones. Tableau handle the datasets very well.. Sharing is simple. User can share work through tableau public, Tableau Server or Tableau cloud.

Data Connections

Tableau help user connect their data sources. It works with Excel file, CSV file , SQL databased.. It also support cloud platform like snowflake and Google BigQuery. Users can choose a connection. This shows the current data. They can use an Extract. An extract saves a copy of data for performance.

In Data Source tab users can join tables link them or join them. This mean data from any source can be brought together. Then all the data can be use for analyzing at once. This flexibility makes Tableau a powerful tool for data work.

Worksheets

A single visualization is built in a worksheet. Users add labels or reference lines. They sort values. They apply filters. They arrange fields on the shelves. Every chart, like a bar chart of sales by region is made in a worksheet. Each worksheet serves as the foundation, for all other work.

Dashboards

A dashboard shows text, pictures, filters and many worksheets on one screen. The dashboard lets you click on one chart to change the charts making the report lively. I like how the dashboard keeps looking nice on both laptops and phones because it uses layout containers and special layouts for each device. Managers can watch performance indicators all in one spot, with the dashboard.

Stories

The narrative arranges dashboards and worksheets into a set of story points each one having a caption. The narrative helps to show facts in a story style, which's good, for management presentations. The narrative starts with the problem shows the evidence and finally gives a recommendation.

Process of creating and customizing visualizations

1. Verify that the fields are correctly categorized by linking to the data source.
2. Move the required measurements and dimensions to the Rows and Columns.
3. Choose the chart type; the "show me" option was helpful.
4. Adjust the Marks card by including filters and sorting once you have made changes, to the color, size, labels and tooltips.
5. Put the worksheets together into a dashboard set the fonts and axes and include a title.

Guiding Principles

The type of chart needs to fit what it is meant to show, like showing parts of something comparing things or showing changes over time., Colors should be used on purpose to highlight parts not to confuse. The design should be easy to understand. The main idea needs to stand out the labels and titles need to be clear.. The numbers should be honest. It is important to think about who will be looking at it.,

In Conclusion, Tableau brings together spreadsheet's dashboards storytelling and easy data connections, in one place. It helps people show data well and make business choices when used with design ideas.

Question 5

Answer:

Advance Data Visualization Technique in Tableau

Beyond charts, Tableau offers advanced features that lead to customized analysis. These include creating calculations by adding interactivity and segmenting data in ways,. These techniques help managers move from looking at data to exploring it and answering specific business questions.

Calculated Fields

A calculated field is a field that is made using a formula based on data that already exists. It can do math check conditions work with text handle dates or group data together. For example Profit Ratio can be made by dividing the profit by the total sales.. An IF statement can also be used to mark an order as 'high' or 'low' based on its value. Tableau calculations and Level of Detail expressions help create running totals percentages of the total or metrics, for each customer no matter what is shown in the view. Calculated fields are important because they let analysts create business KPIs without changing the original data.

Parameters

A parameter is a value that a user can control, such as a number, a date or a list. The parameter can be used in calculations, filters and reference lines.. For example the parameter can let a manager choose between sales,, profit or quantity as the measure shown or set a target value that updates as reference line.

Filters

Filters limit the data that is shown. Tableau offers kinds of filters: extract, data source, context, dimension, measure and table calculation filters. Each filter is applied at a stage of the query order. Users can display menus, sliders or checkboxes. Users can also apply one filter to worksheets. Filters help performance by reducing the amount of data that is processed. Filters keep the analysis focused on what's relevant. Filters can focus on a region, product line or time period.

Sets

A set is a custom subset of data based on a condition. Sets can be fixed, where members are chosen manually, or dynamically, where membership is determined by a rule, such as customer whose sales exceed a certain value or the top ten products by profit. Sets can be compared using IN/OUT, which helps in analyzing top versus other customers. They are useful for cohort analysis, benchmarking, and identifying high-value segments..

Groups

A group brings together related items or a dimension into a higher-level category. For example you can group states into regions. You can merge inconsistent labels like 'Mumbai' and 'Bombay' into one. Groups help reduce clutter in charts and make data easier to read. They clean up the data by removing confusion. Groups are not as flexible, as sets because they depend on manual choices. You have to pick the items yourself when making a group.

Contribution to Effective Analysis

Together these features make visualizations more accurate, engaging and important. Calculated fields create custom measurements; parameters let users have control filters limit the focus and sets and groups put data into parts. They cut down on needing reports allow for more detailed looking, into data and help people who make decisions find insights quicker.

In conclusion calculated fields, parameters, filters, sets and groups change Tableau from a charting tool into a strong analysis tool. Used with design ideas they let people show data clearly and make better-informed business decisions.

Question 6

Answer:

Interactive Dashboards and Storytelling in Tableau

Important Concept and Significance

An interactive dashboard in Tableau is a single screen that combines several worksheets, filters, text and images, which help the users to explore data by just clicking, selecting and filtering of reading a fixed report. Data storytelling goes one step further. Data storytelling arranges these visuals in a sequence so that the data tells a story with a beginning, a middle, and an end. The beginning shows the problem. The middle shows the evidence. The end shows the recommendation.

Significance of dashboard and storytelling for any business is considerable. Managers a overall view point on Key Performance Indicators (KPIs) in one place. Manager can answer follow up questions themselves without waiting for an analyst. Manager can also spot trends and problem areas quickly. Storytelling makes sure that the insights is understood by everyone and acted upon. Decision makers are more likely to remember a narrative than a set of disconnected number.

Filters

Filters let users control the data that appear such as regions, product categories or time frames. In a dashboard a filter can be set up to impact all worksheets so one choice updates every chart at the time. They can look like drop down list, slider or check boxes. Filter help keep the dashboard neat and clean and improve performance by limiting the data that is worked on.

Highlights

Highlighting draws attention to data point while the rest of the data is still there but looks less important. For example picking a product category which makes it stand out across all the graphs helping user to see how it does in every view while still seeing the full picture. This is very useful for looking at and finding trends in measurements

Dashboard Actions

The interactions make a dashboard more interesting. The main types are:

1. **Filter action:** clicking a bar or an area on a map can change the charts
2. **Highlight actions:** Picking a mark makes related data stand out in views.
3. **URL actions:** open a website or an outside report from a selected item.
4. **Sheet actions:** go to another worksheet or dashboard.
5. **Parameter and Set actions:** change a parameter value. Adjust set membership based on what the user wants.

Action let users move from a view, to specific details following the idea of "start with the overview then focus in then get the details when needed."

Multiple Charts

An effective dashboard uses supporting charts like a map for location a line graph for trends a comparison chart for categories and KPIs tiles for important numbers. Each chart can show us the view of same data and together they give a full picture. To keep things, clear the number of charts should be limited the layout should make sense and colors and formatting should be the throughout.

Design and Storytelling Practices

Good dashboards put the important information in the top left use clear titles and labels and make sure colors have a purpose. In Tableau stories each story point has a label that help the people follow the story to an ending.

In Conclusion, Interactive data storytelling change Tableau from just a reporting tool into a place that helps us make better decisions. By using filters, highlights, actions and the right charts analyst can share data that's easy to understand interesting and helpful for making choices.